
CS 350

From Data Mining to Deep Learning

Instructor:
Dr. Peilong Li

Lecture 9:
Clustering

Lecture outline

Announcements/notes

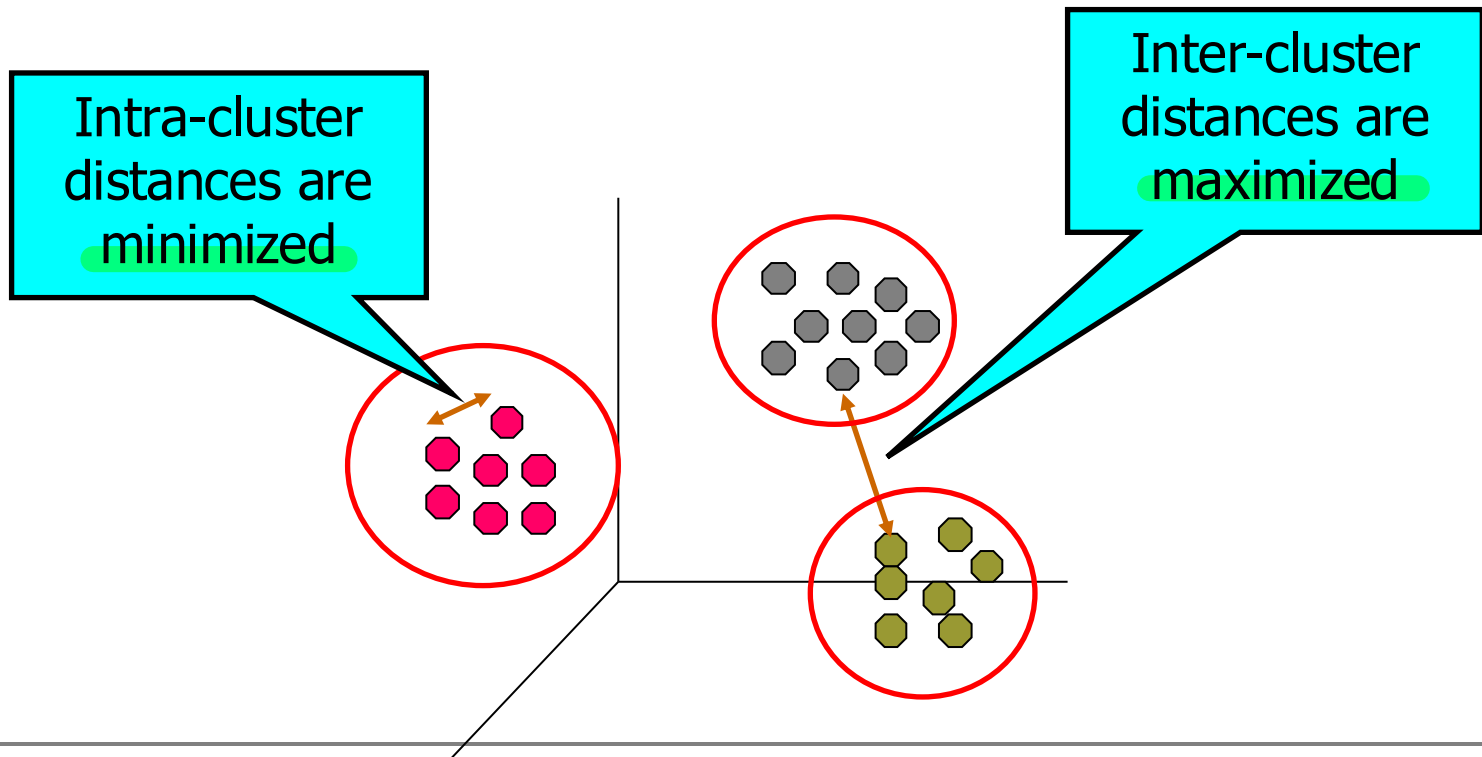
- Read Medium Article for clustering.
- HW 4: Paper Reading is out, due in 2 weeks.
- Project milestone due next Friday.

Today's lecture

- Clustering
 - K-Means
 - Hierarchical Clustering
 - Density-base Clustering

What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Applications of Cluster Analysis

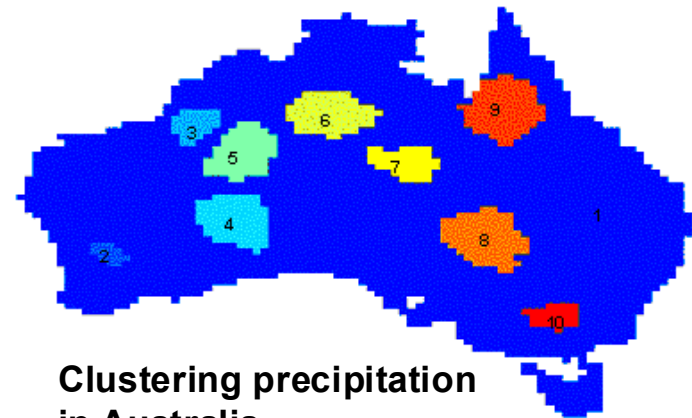
■ Understanding

- Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations

■ Summarization

- Reduce the size of large data sets

	<i>Discovered Clusters</i>	<i>Industry Group</i>
1	Applied-Matl-DOWN,Bay-Network-DOWN,3-COM-DOWN,Cabletron-Sys-DOWN,CISCO-DOWN,HP-DOWN,DSC-Comm-DOWN,INTEL-DOWN,LSI-Logic-DOWN,Micron-Tech-DOWN,Texas-Inst-DOWN,Tellabs-Inc-DOWN,Natl-Semiconduct-DOWN,Oracl-DOWN,SGI-DOWN,Sun-DOWN	Technology1-DOWN
2	Apple-Comp-DOWN,Autodesk-DOWN,DEC-DOWN,ADV-Micro-Device-DOWN,Andrew-Corp-DOWN,Computer-Assoc-DOWN,Circuit-City-DOWN,Compaq-DOWN,EMC-Corp-DOWN,Gen-Inst-DOWN,Motorola-DOWN,Microsoft-DOWN,Scientific-Atl-DOWN	Technology2-DOWN
3	Fannie-Mae-DOWN,Fed-Home-Loan-DOWN,MBNA-Corp-DOWN,Morgan-Stanley-DOWN	Financial-DOWN
4	Baker-Hughes-UP,Dresser-Inds-UP,Halliburton-HLD-UP,Louisiana-Land-UP,Phillips-Petro-UP,Unocal-UP,Schlumberger-UP	Oil-UP



Clustering precipitation
in Australia

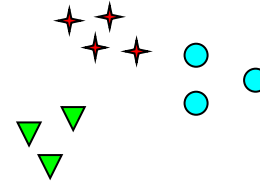
What is not Cluster Analysis?

- Simple segmentation
 - Dividing students into different registration groups alphabetically, by last name
- Results of a query
 - Groupings are a result of an external specification
 - Clustering is a grouping of objects based on the data
- Supervised classification
 - Have class label information
- Association Analysis
 - Local vs. global connections

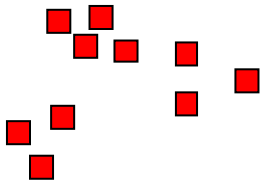
Notion of a Cluster can be Ambiguous



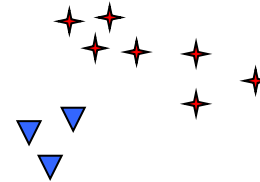
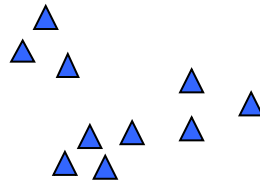
How many clusters?



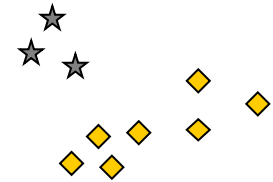
Six Clusters



Two Clusters



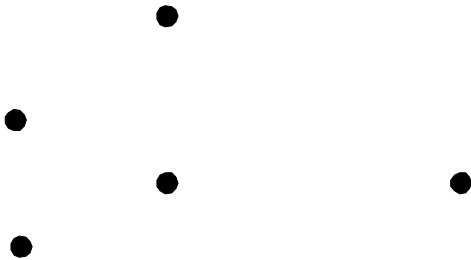
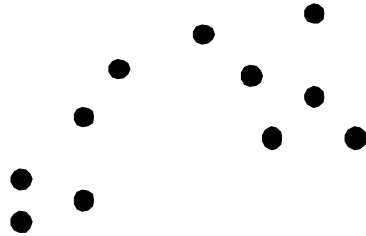
Four Clusters



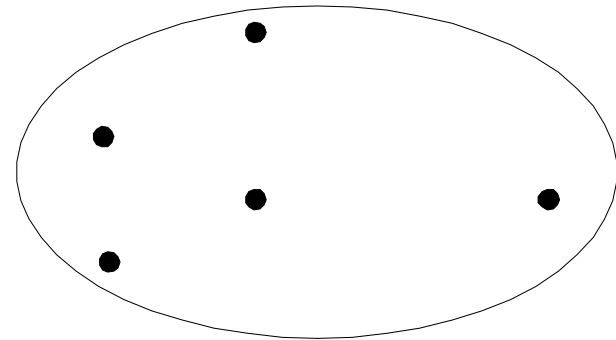
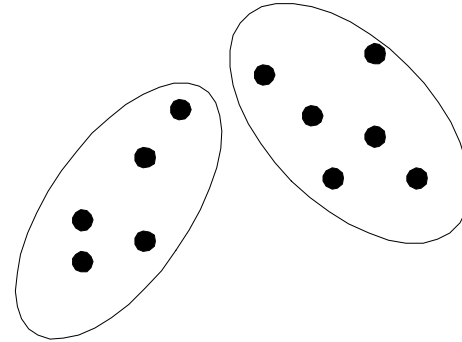
Types of Clusterings

- A **clustering** is a set of clusters
- Important distinction between **hierarchical** and **partitional** sets of clusters
 - Partitional Clustering: A division of data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
 - Hierarchical Clustering: A set of nested clusters organized as a hierarchical tree

Partitional Clustering

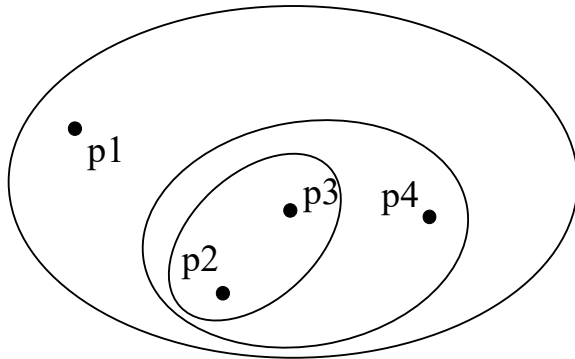


Original Points

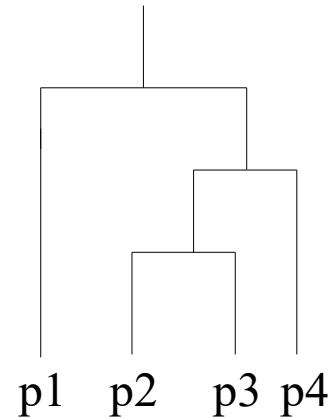


A Partitional Clustering

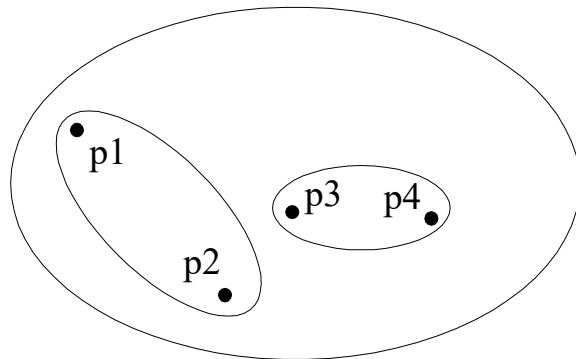
Hierarchical Clustering



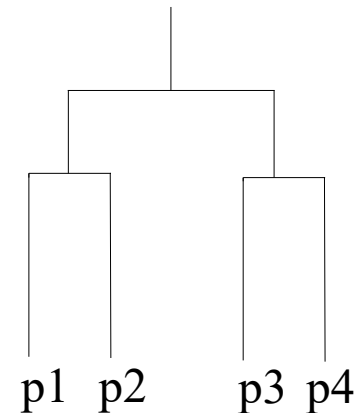
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

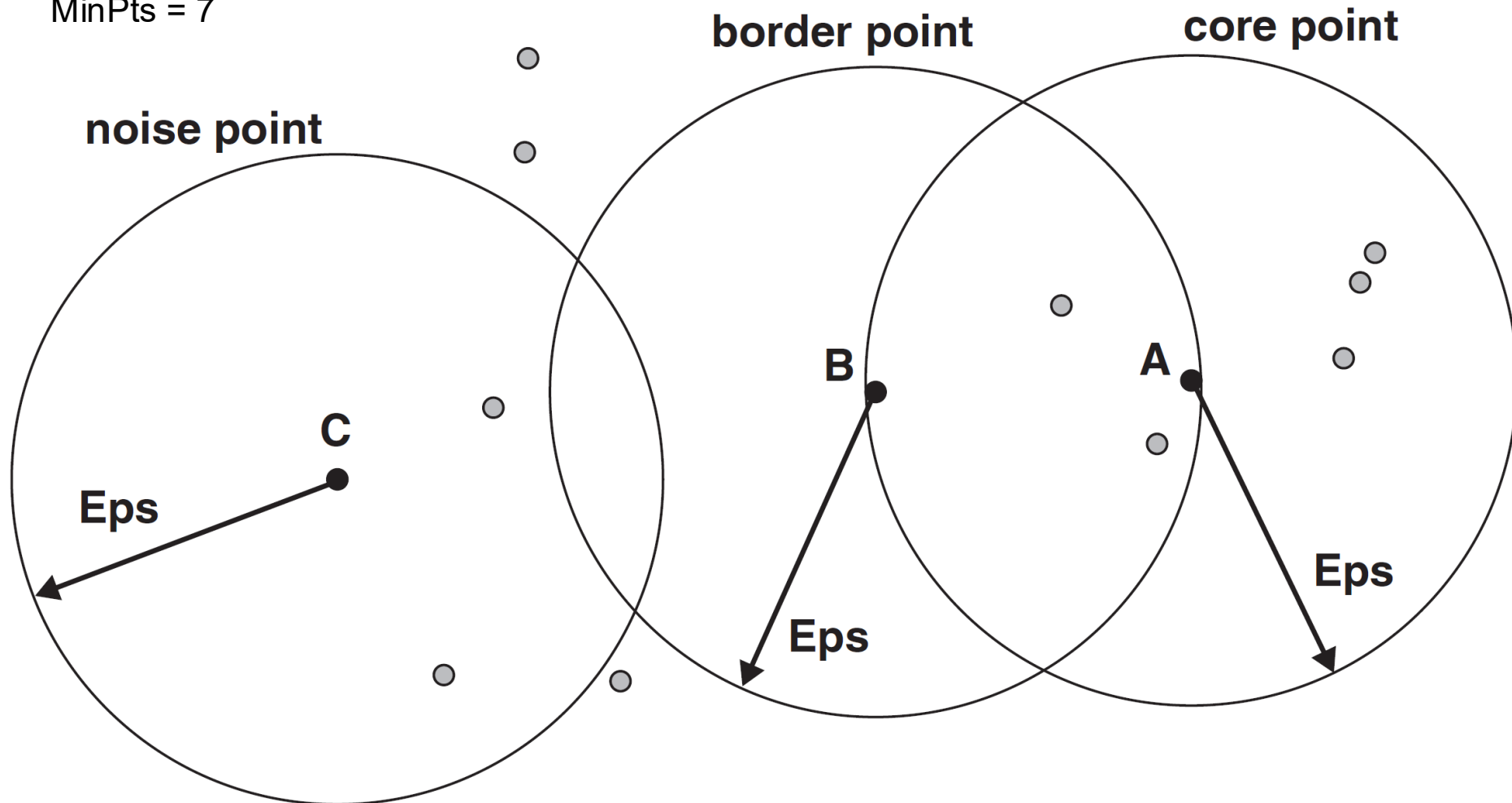
DBSCAN

DBSCAN

- DBSCAN is a density-based algorithm.
 - Density = number of points within a specified radius (Eps)
 - A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
 - These are points that are at the interior of a cluster
 - Counts the point itself
 - A **border point** is not a core point, but is in the neighborhood of a core point
 - A **noise point** is any point that is not a core point or a border point

DBSCAN: Core, Border, and Noise Points

MinPts = 7



DBSCAN Algorithm

- Eliminate noise points
- Perform clustering on the remaining points

$current_cluster_label \leftarrow 1$

for all core points **do**

if the core point has no cluster label **then**

$current_cluster_label \leftarrow current_cluster_label + 1$

 Label the current core point with cluster label $current_cluster_label$

end if

for all points in the Eps -neighborhood, except i^{th} the point itself **do**

if the point does not have a cluster label **then**

 Label the point with cluster label $current_cluster_label$

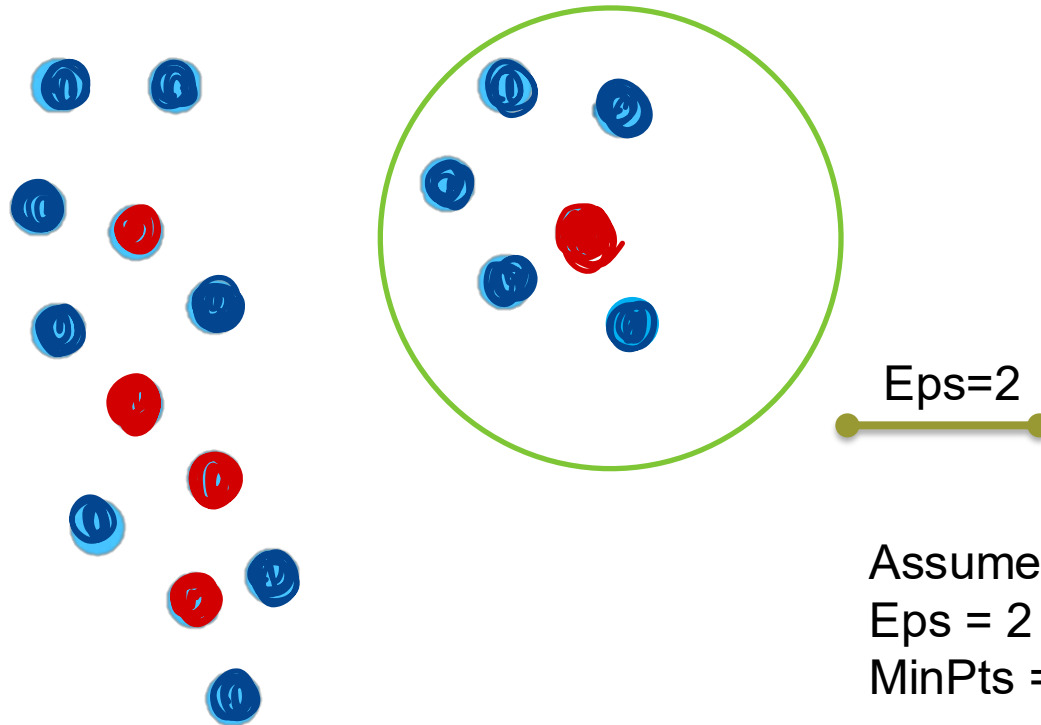
end if

end for

end for

DBSCAN – An Example

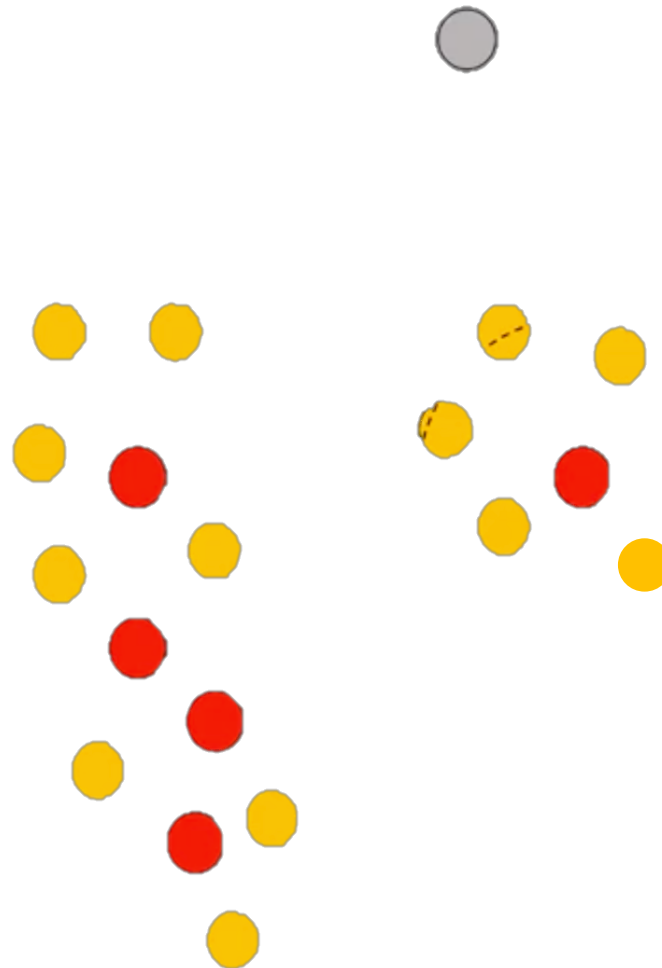
core
border
noise



order
can make
difference

Assume:
Eps = 2
MinPts = 6

DBSCAN – An Example



Red: Core
Yellow: Border
Gray: Outlier

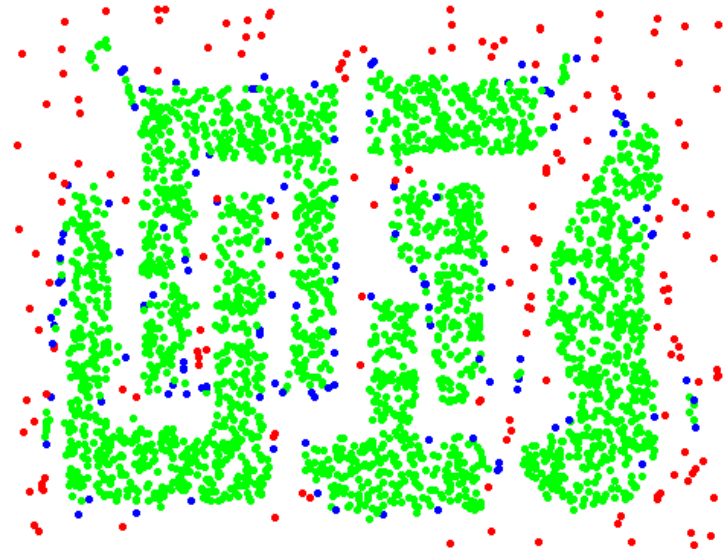
Eps=2

Assume:
Eps = 2
MinPts = 6

DBSCAN: Core, Border and Noise Points



Original Points



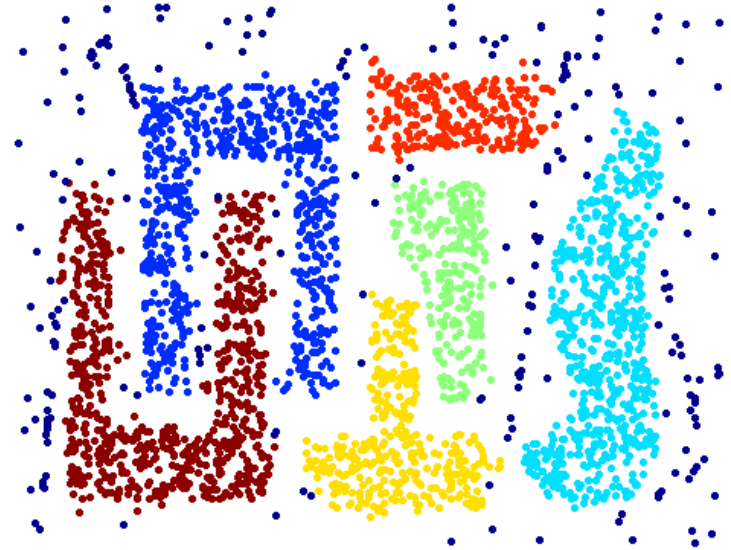
Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

When DBSCAN Works Well



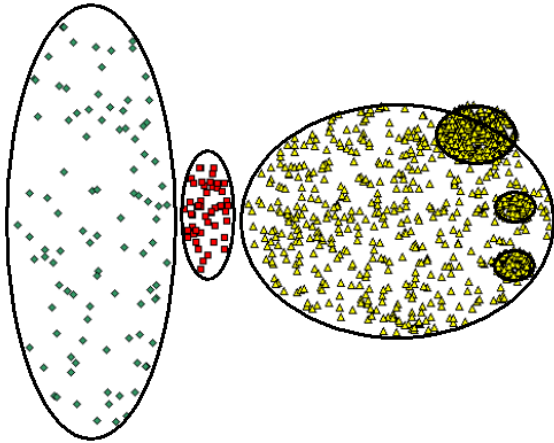
Original Points



Clusters

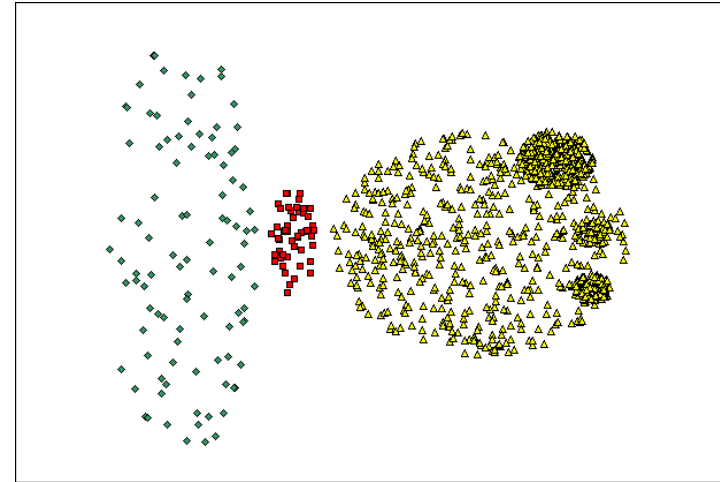
- **Resistant to Noise**
- **Can handle clusters of different shapes and sizes**

When DBSCAN Does NOT Work Well

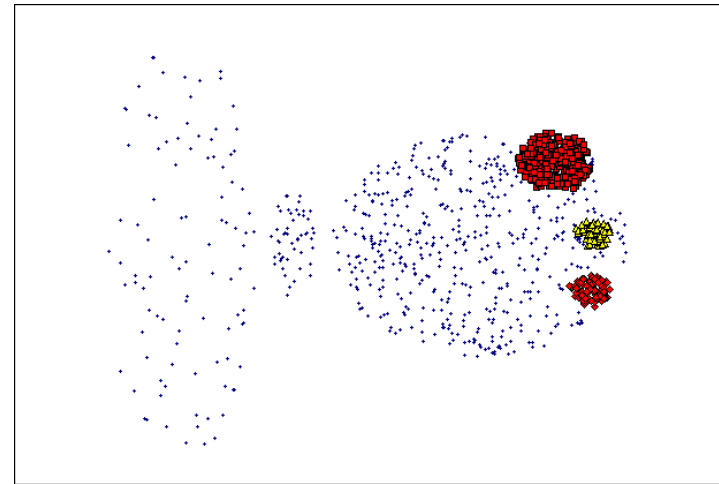


Original Points

- Varying densities
- High-dimensional data



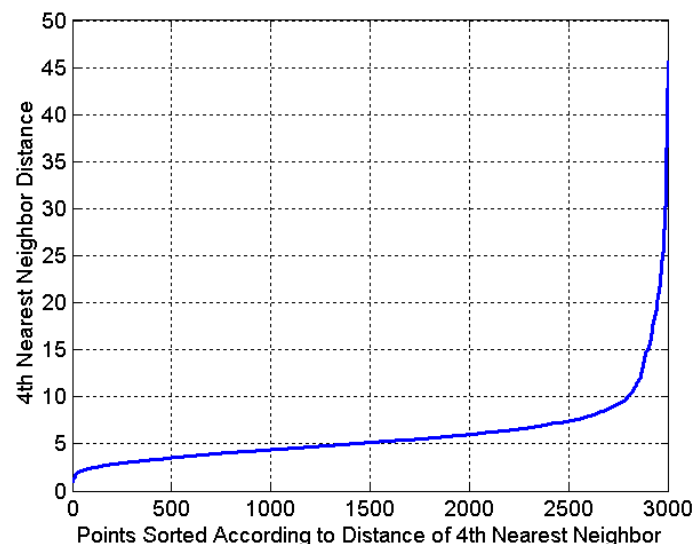
(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their k -th nearest neighbors are at roughly the same distance
- Noise points have the k th nearest neighbor at farther distance
- So, plot sorted distance of every point to its k -th nearest neighbor

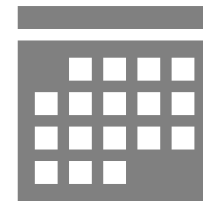


Final notes



Next time:

PE6: Clustering



Reminders:

Read Medium Article

HW4 due in 2 weeks

Project milestone due next Friday